



NMRA Standard	
Train Bus extension Shift Registers (SIO)	
Jan 29, 2026	S-9.4.4 Draft

1 General

1.1 Introduction and Intended Use (Informative)

This standard describes an interface between a main module, hereinafter referred to as Host, and shift register extension modules, hereinafter referred to as SIO-Modules, installed in the model railway vehicles. These extension modules can be used for controlling additional functions that does not require programmed components. This makes easy the controlling of more functions than the number of connections on the decoder interface allows.

The control and assignment of the functions is completely under the control of the decoder.

This is an alternative use of the Train Bus connections, e.g., according to [S-9.4.1]. Simultaneous use with other functional extensions such as SUSI is therefore not possible.

1.2 References

This standard should be interpreted in the context of the following NMRA Standards, Technical Notes, and Technical Information.

1.2.1 Normative

When using the corresponding interface, the following standards must be complied:

- [S-9.4.1] SUSI bus module expansion interface

1.2.2 Informative

The standards and documents listed here are for information only and are not part of this standard.

NONE

1.3 Terminology

Term	Definition
HOST	Main module which generate the SIO DATA and CLOCK signals
SIO-Module	Extension module, controlled by the Host

1.4 Requirements

To be compliant with this standard, no other standards must be respected (followed).

2 Electrical properties

2.1 General electrical properties

Recommended cable colors for the 4-pin interfaces:

1 Ground	Black
2 Data	Gray (SIO Data connected to the Train Bus Data signal of the decoder)
3 Clock	Blue (SIO Clock connected to the Train Bus Clock signal of the decoder)
4 V+	Red

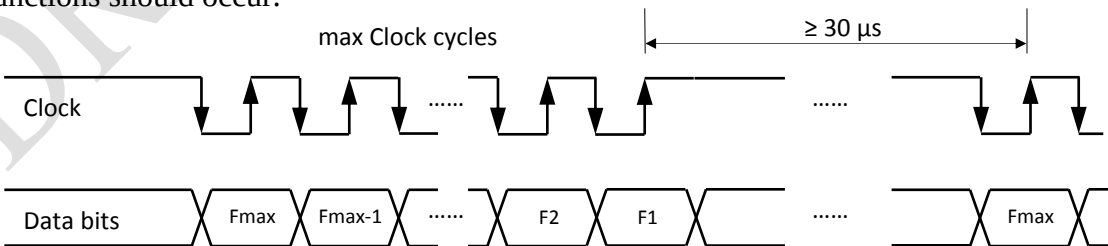
2.2 Logic levels at the interface

The circuit input is connected to the Train Bus terminals of the corresponding interface. The levels specified in the interface standards must be respected:

	Decoder output	Shift register input
Low level	≤ 0.4 Volt	≤ 0.8 Volt
High level	≥ 2.4 Volt	≥ 2.0 Volt

Table 1: Voltage level at the input of the shift register circuit.

- The Data is transferred to the shift register on the positive edge of the Clock signal.
- The shift Clock must have a frequency at least 400 kHz and not more than 4 MHz.
- To transfer the Data, the Clock must be interrupted for at least 30 μ s.
- The length of the shift register can have maximum 16 bits.
- The highest function Data bit is always transmitted on the first Clock, the lowest function Data bit is always transmitted on the last Clock. In this way, if more or fewer Clock cycles are transmitted than required, due to incorrect configuration of the decoder, no offset in the lower functions should occur.



The shift Clock should be configurable in the decoder to adapt to different cable lengths or have a minimum value.

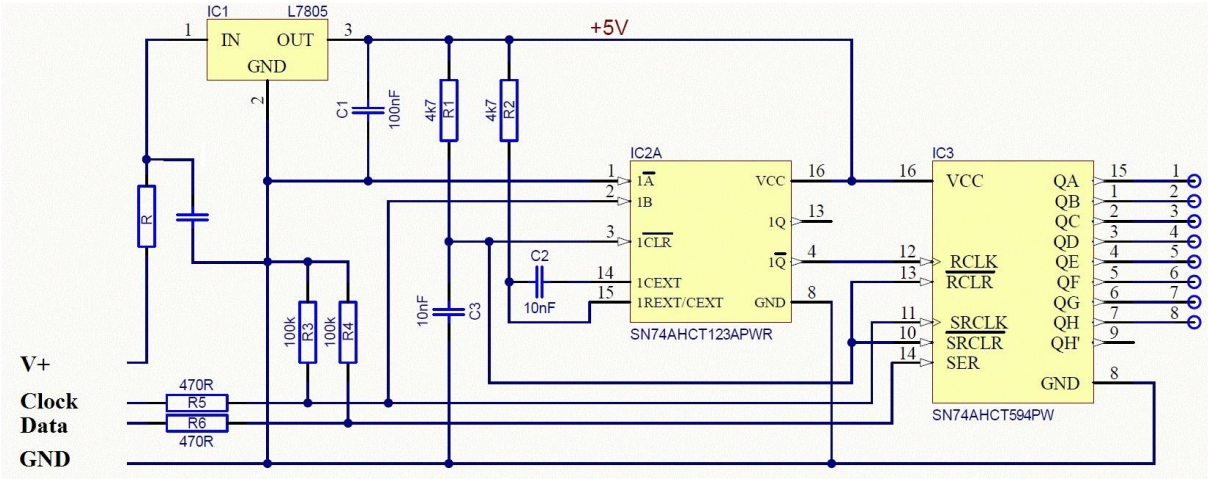


Figure 1: Circuit example with one single 8-bit shift register

4 Document History

Date	Description
2025.11.14	First Draft
2025.12.18	Fraction separator corrected, Clock diagram included, text reformulated in line 80
2026.01.29	Reordered the “Document History” and “Table of Contents” chapters.

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